

1. (20 points) **Measuring the Cosmological Constant in the Laboratory:** Measuring the Cosmological Constant in the Laboratory: It's not possible to measure the matter density  $\rho_M$  in a laboratory of the typical size found on Earth because the FRW approximation that the matter is smoothly distributed breaks down at those scales. However, a fundamental vacuum energy could be exactly uniform and therefore in principle detectable in a laboratory experiment. Calculate how two test particles in a freely falling laboratory would move relative to each other in the presence of a vacuum energy corresponding to  $\Omega_\Lambda = 1$ . Estimate the time scale for significant relative motion and the size of their relative acceleration assuming they start 1 cm apart. Is laboratory detection feasible?
2. **Quantum Gravity:** Could the observed vacuum mass-energy density in the universe be a consequence of quantum gravity? One obstacle to such an explanation is the great difference in scale between observed vacuum mass density  $\rho_\Lambda$  and the Planck mass density  $\rho_{\text{Pl}} \equiv c^5/\hbar G^2$  that might be expected on dimensional grounds to characterize quantum gravitational phenomena.
  - (a) (5 points) Show that  $\rho_{\text{Pl}}$  is the correct combination of  $\hbar$ ,  $G$ , and  $c$  with the dimensions of mass density.
  - (b) (10 points) Evaluate the ratio  $\rho_\Lambda/\rho_{\text{Pl}}$ .
3. In cosmology we tend to idealize nonrelativistic particles as having zero temperature  $T$  and pressure  $p$ . In reality, random motions will give them some temperature and pressure, satisfying  $p \propto T\rho$ .
  - (a) (5 points) How does the pressure of a gas of massive particles decay as a function of the scale factor?
  - (b) (10 points) Suppose neutrinos have a mass  $m_\nu = 0.1$  eV and a current temperature  $T_{\nu 0} = 2$  K. At about what redshift did the neutrinos go from being relativistic to nonrelativistic?